QUASI-COHERENCE FOR THE CHAOTIC TOPOLOGY: SECTIONS OF A QUASI-COHERENT MODULE FORM A BASE-CHANGED MODULE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two external items (an example and the definition), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- The chaotic (indiscrete) topology and its coverings. Stacks Project, Sites, Example sites-example-indiscrete (tag 00V8). Gives that coverings are isomorphisms, which makes the comparison with module theory possible.
- Definition of quasi-coherence on a ringed site. Stacks Project, Modules on Sites, Definition definition-site-local (tag 09B9). Supplies the presentation the proof starts from and reconstructs at the end.

Source reference. Original reference 07GE: sources/stacks-chow/tags.txt, source lines 9514–9514. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a category viewed as a site with the chaotic topology, see Sites, Example 07GE. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}$ and let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Then $\mathcal{F}$ is quasi-coherent if and only if for all $U \to V$ in $\mathcal{C}$ the canonical map*

$$\mathcal{F}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) \longrightarrow \mathcal{F}(U)$$

is an isomorphism.

Proof. Assume $\mathcal{F}$ is quasi-coherent and let $U \to V$ be a morphism of $\mathcal{C}$. Since every covering of V is given by an isomorphism we conclude from Definition 03DL that there exists a presentation

$$\bigoplus\nolimits_{j \in J} \mathcal{O}_V \longrightarrow \bigoplus\nolimits_{i \in I} \mathcal{O}_V \longrightarrow \mathcal{F}|_{\mathcal{C}/V} \longrightarrow 0$$

Since the topology on $\mathcal{C}$ is chaotic, taking sections over any object of $\mathcal{C}$ is exact. We conclude that we obtain a presentation

$$\bigoplus\nolimits_{j \in J} \mathcal{O}(V) \longrightarrow \bigoplus\nolimits_{i \in I} \mathcal{O}(V) \longrightarrow \mathcal{F}(V) \longrightarrow 0$$

of $\mathcal{F}(V)$ as an $\mathcal{O}(V)$-module and similarly for $\mathcal{F}(U)$. This easily shows that the displayed map in the statement of the lemma is an isomorphism.

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Assume the displayed map in the statement of the lemma is an isomorphism for every morphism $U \to V$ in $\mathcal{C}$. Fix V and choose a presentation

$$\bigoplus\nolimits_{j \in J} \mathcal{O}(V) \longrightarrow \bigoplus\nolimits_{i \in I} \mathcal{O}(V) \longrightarrow \mathcal{F}(V) \longrightarrow 0$$

of $\mathcal{F}(V)$ as an $\mathcal{O}(V)$-module. Then the assumption on $\mathcal{F}$ exactly means that the corresponding sequence

$$\bigoplus\nolimits_{j \in J} \mathcal{O}_V \longrightarrow \bigoplus\nolimits_{i \in I} \mathcal{O}_V \longrightarrow \mathcal{F}|_{\mathcal{C}/V} \longrightarrow 0$$

is exact and we conclude that $\mathcal{F}$ is quasi-coherent. $\square$